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3) Discuss the importance and role of the Capital Pricing Model in an economy.

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4) List some applications of the Capital Pricing Model.

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19.5 LIMITATION OF CAPITAL ASSET PRICING MODEL

Capital Asset Pricing Model (CAPM) is widely used and provides valuable insights into the relationship between risk and return in financial markets, it also has several limitations. Some of the key limitations include:

1) **Assumptions:** CAPM is based on several simplifying assumptions that may not always hold in real-world financial markets. These assumptions include perfect competition, frictionless markets, constant correlations among assets, and rational investor behaviour. Deviations from these assumptions can lead to inaccuracies in CAPM's predictions.

2) **Market Index:** CAPM relies on the market index to represent the overall market portfolio. However, the choice of the market index can significantly impact the results. Different market indices may have different compositions and characteristics, leading to variations in beta estimates and expected returns.

3) **Single-Factor Model**: CAPM is a single-factor model that considers only systematic risk (beta) as the sole determinant of expected returns. It does not account for other factors such as firm-specific risk, macroeconomic variables, or market sentiment, which can also influence asset prices and returns. As a result, CAPM may not fully capture the complexities of real-world asset pricing.

4) **Linearity and Homogeneity:** CAPM assumes linear and homogeneous relationships between risk and return, implying that all investors have the same risk preferences and risk aversion coefficients. In reality, investors

have diverse risk preferences and may exhibit nonlinear behaviour, leading to deviations from CAPM predictions.

5) **Risk-Free Rate:** CAPM relies on the risk-free rate as a proxy for the time value of money and the opportunity cost of capital. However, the risk-free rate may fluctuate over time and may not accurately reflect investors' actual borrowing and lending opportunities. Changes in the risk-free rate can impact the entire risk-return relationship predicted by CAPM.

6) **Empirical Validity:** Empirical studies have found mixed evidence regarding the validity of CAPM in explaining asset returns. While CAPM provides a useful benchmark for asset pricing, its ability to accurately predict returns in practice has been questioned, especially during periods of market turbulence or financial crises.

7) **Non-Market Risks:** CAPM focuses primarily on market (systematic) risk and may overlook non-market (idiosyncratic) risks that are specific to individual assets or industries. These non-market risks, such as company-specific events or regulatory changes, can significantly impact asset prices and returns but are not captured by CAPM.

Overall, while CAPM remains a foundational model in financial economics, it is essential to recognize its limitations and consider alternative models or factors when making investment decisions and assessing asset pricing in real-world financial markets.

19.5.1 Theoretical Limitation of Capital Asset Pricing Model

The theoretical limitation of the Capital Asset Pricing Model (CAPM) lies in its assumptions, which may not always hold in real-world financial markets. Some of the limitations are discussed as follows:

1) **Perfect Market Assumptions:** CAPM assumes the existence of perfect markets, where investors have access to all relevant information, transaction costs are zero, and there are no restrictions on borrowing or lending. In reality, markets are often imperfect, with information asymmetries, transaction costs, and regulatory constraints affecting investor behaviour and asset prices. For example: During the global financial crisis of 2007-2008, financial markets experienced significant disruptions, with widespread information asymmetries, liquidity shortages, and regulatory interventions. These market imperfections challenged the assumptions of perfect markets underlying CAPM, leading to distortions in asset pricing and deviations from CAPM predictions.

2) **Homogeneous Investor Behaviour:** CAPM assumes that all investors have homogeneous expectations, risk preferences, and investment horizons. In reality, investors exhibit diverse behaviour, with varying

risk aversion levels, investment goals, and time horizons. This heterogeneity in investor behaviour can lead to discrepancies between CAPM-predicted asset prices and actual market prices. For Example: During periods of market volatility, investors may exhibit different risk preferences and reactions to market events. Some investors may adopt a more risk-averse stance and demand higher returns for bearing risk, while others may be more willing to take on risk in pursuit of higher returns. These heterogeneous behaviours can lead to deviations from the CAPM-predicted relationship between risk and return.

3) **Single-Factor Model:** CAPM is a single-factor model that considers only systematic risk (beta) as the sole determinant of expected returns. It does not account for other factors such as firm-specific risk, macroeconomic variables, or market sentiment, which can also influence asset prices and returns. This limitation restricts CAPM's ability to fully capture the complexities of asset pricing dynamics. For Example, The financial markets often experience periods of heightened uncertainty driven by macroeconomic events, geopolitical tensions, or changes in investor sentiment. During such periods, factors beyond systematic risk, such as changes in interest rates, inflation expectations, or regulatory policies, can exert significant influence on asset prices, leading to deviations from CAPM predictions.

These theoretical limitations highlight the challenges in applying CAPM in real-world financial markets and underscore the importance of considering alternative models or factors to better understand asset pricing dynamics and make informed investment decisions.

19.5.2 Practical Limitation of Capital Asset Pricing Model

The Capital Asset Pricing Model (CAPM) is a valuable tool in finance, but it does have practical limitations.

1) **Non-Linear Relationship between Risk and Return:** CAPM assumes a linear relationship between risk (measured by beta) and expected return. However, in reality, the relationship between risk and return may not always be linear. Assets with higher betas may not always provide proportionally higher returns, and the relationship may exhibit non-linearity, especially at extreme ends of the risk spectrum. This limitation can be explained with the help of an example of a Tech Start-Up. Consider the case of investing in technology startups. These startups often exhibit high levels of risk due to factors such as uncertain revenue streams, competition, and technological disruption. According to CAPM, investors would expect higher returns for investing in these high-risk ventures. However, the relationship between risk and return may not be strictly linear. While some successful startups may provide significant returns, many others may fail, leading to losses for investors despite their

high-risk nature. This non-linear relationship challenges CAPM's ability to accurately predict returns for such high-risk investments.

2) **Sensitivity to the inputs and assumptions used in its calculations** can lead to unreliable estimates of expected returns. CAPM relies on several inputs and assumptions, including the risk-free rate, market risk premium, and beta coefficients. However, these inputs are subject to uncertainty and may vary over time, leading to potential inaccuracies in CAPM's predictions. Small changes in these inputs can result in significant fluctuations in estimated expected returns, reducing the reliability of CAPM as a forecasting tool. This limitation is also explained with the help of the following example of Sensitivity to market risk and premium. Consider the estimation of the market risk premium, which represents the excess return investors expect from investing in the market portfolio compared to the risk-free rate. This parameter is crucial in determining expected returns according to CAPM. However, the market risk premium is not directly observable and must be estimated based on historical data or other methodologies. If different analysts or investors use different methods or data sources to estimate the market risk premium, they may arrive at divergent estimates. Additionally, the market risk premium itself may fluctuate over time due to changes in economic conditions, investor sentiment, or market dynamics. As a result, small variations in the estimated market risk premium can lead to significant differences in expected returns according to CAPM. The sensitivity of CAPM to its inputs and assumptions can have practical implications for investment decision-making. Investors and financial analysts may struggle to obtain accurate estimates of expected returns using CAPM, especially in dynamic or uncertain market environments. Relying solely on CAPM's predictions without considering the inherent uncertainty and sensitivity of its inputs can result in suboptimal investment decisions and portfolio performance.

3) **Poor predictor of returns:** Empirical studies support that the reliability of CAPM becomes weak when the estimate of asset returns is closely associated with realized returns. Empirical findings do not support the hypothesis that asset returns are only determined by systematic risk. The low predictive power of the CAPM model is one of the serious limitations because investors invest based on a prediction of future returns.

Check Your Progress 2

1) Discuss the limitations of the Capital Asset Pricing Model.

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2) Describe the following:

   (a) Theoretical Limitation of Capital Asset Pricing Model.

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   (b) A practical limitation of the Capital Asset Pricing Model.

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19.6 EMPIRICAL ANALYSIS OF THE CAPM MODEL

The Capital Asset Pricing Model (CAPM) can be applied to real-world financial data and the results can be tested with its predictions. Some of the applications of the CAPM model are as follows:

Time-Series Analysis: Economists collect historical data on individual assets' returns and the overall market return over a specified period. Using regression analysis, they estimate the beta coefficients for each asset to measure their sensitivity to market movements. CAPM's predictions are then tested by comparing the expected returns calculated using CAPM with the actual historical returns of the assets. Example: A study examines the relationship between the beta coefficients of different stocks and their subsequent returns over 10 years. Economists use CAPM to estimate the expected returns and assess whether they closely match the actual returns observed in the market.

Cross-Sectional Analysis: Economists examine the relationship between asset returns and their beta coefficients across different assets at a specific point in time. Using statistical tests, they assess whether there is a significant positive relationship between beta and expected returns, as predicted by CAPM. An example: A study analyzes a sample of stocks from different industries and calculates the beta coefficients for each stock. Researchers then compare the expected returns derived from CAPM with the actual returns of the stocks to determine if CAPM holds across a diverse set of assets.

Event Studies: Researchers conduct event studies to examine the impact of specific events, such as earnings announcements or mergers, on asset returns. They use CAPM to predict how asset prices should respond to these events based on their beta coefficients and market returns.

Example: A study investigates the stock market's reaction to quarterly earnings announcements. Researchers calculate the expected returns using

CAPM and compare them with the actual returns observed around the announcement dates to evaluate CAPM's ability to explain asset price movements.

Portfolio Performance Evaluation: Investors use CAPM to assess the performance of their investment portfolios relative to their level of systematic risk. They calculate the portfolio's beta and compare its expected return, as predicted by CAPM, with the actual return achieved by the portfolio. Example: An investor analyzes the performance of their diversified portfolio of stocks over the past year. They use CAPM to estimate the expected return of the portfolio based on its beta and compare it with the actual return to evaluate whether the portfolio outperformed or underperformed relative to its level of risk.

Let us illustrate how empirical analysis of CAPM can be conducted using different methodologies and datasets to assess the model's validity and applicability in real-world financial markets. Suppose in a market:

- Risk-free rate (R_f): 3%
- Market return ($E(R_m)$): 8%
- Stock A's beta (β_A): 1.2

Using CAPM, we can calculate the expected return ($E\ (R_A)$) for Stock A as follows:

$E(R_A)=R_f+\beta_A\times(E(R_m)-R_f)$

Substituting the given values:

=0.03+1.2× (0.08−0.03) E (RA) =0.03+1.2× (0.08−0.03)

=0.03+1.2×0.05E (RA) =0.03+1.2×0.05

=0.03+0.06E (RA) =0.03+0.06

=0.09E (RA) =0.09

So, according to CAPM, the expected return for Stock A is 9%.

Now, let's assume that the actual return for Stock A over a specific period was 10%. We can then assess whether this actual return aligns with the expected return predicted by CAPM.

We can compare the actual return with the expected return:

In this case, the Actual return i.e., 10% is greater than the Expected return i.e., 9%. Since the actual return for Stock A is greater than the expected return calculated using CAPM, this suggests that Stock A outperformed relative to its level of systematic risk, as predicted by CAPM.

19.7 LET US SUM UP

CAPM is a fundamental model in finance that establishes a relationship between risk and return for individual assets or portfolios in a well-diversified market.

CAPM focuses on systematic risk, which cannot be diversified away, and measures it using beta (β), representing the asset's sensitivity to market movements.

According to CAPM, the expected return of an asset is determined by the risk-free rate (Rf) plus a risk premium proportional to the asset's beta and the market risk premium (E(Rm)−Rf).

The formula for expected return E(R)) using CAPM is: $E(R) = Rf + \beta \times (E(Rm) - Rf)$

A risk-free rate represents the return on a risk-free investment, typically proxied by government bonds, with no default risk.

The excess return investors demand for holding a risky asset compared to a risk-free asset. It is calculated as the difference between the expected return of the market portfolio E(Rm)) and the risk-free rate (Rf).

A theoretical portfolio that contains all assets in the market, weighted by their market values. It represents the average level of risk in the market.

CAPM implies that all investors will hold a combination of the risk-free asset and the market portfolio, known as the efficient frontier, to achieve optimal risk-return trade-offs. It relies on several assumptions, including perfect capital markets, rational investors, homogeneous expectations, and constant risk aversion. CAPM is widely used in finance for asset pricing, portfolio management, and performance evaluation. It provides a benchmark for determining whether an asset's expected return adequately compensates for its level of risk.

Despite its widespread use, CAPM has limitations, including its simplifying assumptions, reliance on historical data, inability to capture non-market risks, and mixed empirical evidence.

Researchers have developed alternative models and extensions to CAPM, such as multi-factor models and behavioural finance frameworks, to address its limitations and improve the understanding of asset pricing dynamics.

To conclude, CAPM serves as a foundational framework for understanding the relationship between risk and return in financial markets, guiding investment decisions and portfolio management strategies. However, it is essential to recognize its assumptions and limitations when applying them in real-world contexts.

19.8 KEY WORDS

Anomalies	:	Patterns or inconsistencies in asset returns that are not explained by CAPM.
Asset Pricing	:	The process of valuing assets based on their risk-return characteristics.
Beta	:	The measure of an asset's sensitivity to market movements.
Diversification	:	Spreading investments across different assets to reduce risk.
Efficient Frontier	:	The set of optimal portfolios that offer the highest expected return for a given level of risk.
Empirical Analysis	:	Examination of real-world data to test the validity and applicability of economic theories like CAPM.
Expected Return	:	The return an investor anticipates receiving on an investment is based on its risk characteristics.
Market Efficiency	:	The degree to which prices of assets accurately reflect all available information in the market.
Market Portfolio	:	A theoretical portfolio containing all assets in the market, weighted by their market values.
Market Risk Premium	:	Excess return expected from holding a risky asset compared to risk-free asset.
Portfolio Theory	:	The study of how investors can construct portfolios to optimize returns while minimizing risk.
Return	:	The gain or loss generated on an investment over a specific period.
Risk	:	The variability of returns associated with an investment.
Risk-Free Rate	:	The return on a risk-free investment, is usually proxied by government bonds.
Systematic Risk	:	Market-related risks that cannot be diversified away

19.9 SOME USEFUL BOOKS

Abel B (1990) "Asset Prices under Habit Formation and catching up with the Jones; The American Review 80(2) pp 38-42

Chen N Roll and Ross S (1986) “Economic forces and the Stock Market Journal of Business 59(3) pp 383-403

Frank J Faboozi (2006) “Financial Modeling of the Equity Market from CAPM to Cointegration.

Haim Levy (2011) The Capital Asset Pricing Model in the 21st Century

Martin L and Gonzalo R (2011) Evaluating alternative methods forecasting asset pricing models with historical data. “Journal of Empirical Finance 18(1) pp 136-146

19.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) CAPM provides a theoretical foundation for understanding the relationship between risk and return in financial markets.

2) Emphasis on systematic risk, single-factor model, relationship between risk and return, market portfolio benchmark, risk-free rate incorporation, equilibrium pricing, universal applicability.

3) Objective measurement of risk-adjusted performance, quantifying risk-return tradeoff, portfolio allocation decision, performance evaluation tool, risk management tool, benchmark comparison.

4) Expected return estimation, cost of equity calculation, risk-adjusted performance measurement.

Check Your Progress 2

1) Strict assumptions, market index, single-factor model, linearity and homogeneity, risk-free rate, empirical validity, non-market risks.

2) a) Perfect Market Assumptions, Homogeneous Investor Behaviour, Single-Factor Model.

   b) Non-linear relationship between risk and return, sensitivity to the inputs and assumptions used in its calculations, poor predictor of returns.

19.11 TERMINAL QUESTIONS

1) What is the Capital Asset Pricing Model (CAPM), and what are its key assumptions?

2) Provide the CAPM equation and explain the significance of each component.

3) Discuss common statistical measures used to evaluate the goodness-of-fit of CAPM in empirical analyses.

4) Explain how the presence of anomalies affects the applicability of CAPM in real-world financial markets.

5) Discuss the theoretical and practical limitations of CAPM.